A MORPHISM FACTORS UNIQUELY THROUGH ITS COIMAGE AND ITS IMAGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument, including the uniqueness of the middle map and the vanishing of the composite into the cokernel; no step is left to the reader and the closing sentence is a conclusion.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- Images, coimages, kernels and cokernels. Same chapter, the definitions of the kernel, cokernel, image and coimage of a morphism in a preadditive category. Supplies the objects and the universal properties through which the two factorisations in the proof are produced.
- A kernel map can be cancelled on the left. Same chapter, Lemma lemma-kernel-mono. Supplies the uniqueness statement used to conclude that the composite into the cokernel is zero.

Source reference. Original lemma 0E43: sources/stacks-homology/pinned-original-source.tex, source lines 492–492. Navigation pointer invoked at proof line 530; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $f : x \to y$ be a morphism in a preadditive category such that the kernel, cokernel, image and coimage all exist. Then f can be factored uniquely as $x \to \mathrm{Coim}(f) \to \mathrm{Im}(f) \to y$.*

Date: stacks-project:homology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. There is a canonical morphism $\mathrm{Coim}(f) \to y$ because $\mathrm{Ker}(f) \to x \to y$ is zero. The composition $\mathrm{Coim}(f) \to y \to \mathrm{Coker}(f)$ is zero, because it is the unique morphism which gives rise to the morphism $x \to y \to \mathrm{Coker}(f)$ which is zero (the uniqueness follows from Lemma 0E43 (3)). Hence $\mathrm{Coim}(f) \to y$ factors uniquely through $\mathrm{Im}(f) \to y$, which gives us the desired map. $\square$